Embedded Systems Development

Exercise 4

1. Design an "auto light" using ATmega328P microcontroller and give your design in a schematic diagram and a wiring diagram.

   The "auto light" is an embedded system that turns on and off an LED depending on the ambient light intensity. It captures ambient light intensity using a transducer and automatically turns on an LED at the nightfall and turns it off in the morning light. The "auto light" explains how the variation of a physical phenomenon is captured and converted into a voltage using a transducer, maps the voltage into a number so that a CPU can process and then produce a digital output to drive an actuator.

   Schematic Diagram

Wiring Diagram

2. Write the required firmware for the "auto light" using wrapper functions in Arduino IDE.

3. Write the required firmware for the "auto light" in C.

4. Extend your "auto light" design given in (1) above to handle the turning on and off of a light connected to the domestic power supply (230V A/C) and give your design in a schematic diagram and a wiring diagram.



Warning - This extension of the circuit involves in operating at a High Voltage with a potential for giving a fatal shock. Therefore, do **NOT** practically experiment it unless you really know what you are doing at your own risk.

Schematic Diagram

Wiring Diagram